

KRegCCD: the 15 four-taxon tree probabilities, symbolically

Setup

Taxa $\{A, B, C, D\}$. Two trees observed, each once:

$$T_1 = (((A, B), C), D), \quad T_2 = (((D, C), B), A),$$

so the observed non-trivial clades are $\{A, B\}, \{A, B, C\}$ (from T_1) and $\{C, D\}, \{B, C, D\}$ (from T_2). Clades never seen are the **blue nodes**. KRegCCD is fitted with smoothing parameter α and per-clade escape probability μ (reserve depth $k = 2$, `NovelMode.FLAT`). With four taxa there are exactly $(2 \cdot 4 - 3)!! = 15$ rooted topologies, so this is the whole tree space.

The three building blocks

The escape probability μ is a single global parameter: every reservable clade C reserves the *same* total mass μ for its blue resolutions. The escape root $\varepsilon(C)$, by contrast, is solved *per clade* from that clade's own reserve equation

$$R(C) = \sum_{j \geq 1} N_j(C) \varepsilon(C)^j = \mu,$$

where $N_j(C)$ counts the depth- j novel resolutions of C . So $\varepsilon(C)$ varies from clade to clade (the root and the 3-taxon clades below get different roots), while μ does not. The two distinct values $\varepsilon_{\text{root}}$ and ε_3 used in the table are $\varepsilon(C)$ at the root and at a 3-taxon clade respectively.

Backbone CCPs (from α). Additive- α smoothing at the root clade $\{A, B, C, D\}$, which was seen twice and carries three candidate splits (the two observed splits plus the split-expanded balanced split $\{A, B\}|\{C, D\}$), gives

$$c_{\text{obs}} = \frac{1 + \alpha}{2 + 3\alpha} \quad (\text{each observed split}), \quad c_{\text{bal}} = \frac{\alpha}{2 + 3\alpha} \quad (\text{balanced split}),$$

with the key identity $2c_{\text{obs}} + c_{\text{bal}} = 1$. The two interior 3-taxon clades each carry a single split, so their CCP is 1.

Root escape root $\varepsilon_{\text{root}}$. The root reserves total mass μ for its blue resolutions. In FLAT mode the reserve sums tree counts: $M_1 = 2$ trees carry one novel clade and $M_2 = 6$ carry two, so the root's escape root $\varepsilon_{\text{root}}$ solves

$$2\varepsilon_{\text{root}} + 6\varepsilon_{\text{root}}^2 = \mu \quad \implies \quad \varepsilon_{\text{root}} = \frac{\sqrt{1 + 6\mu} - 1}{6}.$$

Interior 3-clade escape root $\varepsilon_3 \equiv \varepsilon(\mathbf{3-clade})$. A 3-taxon observed clade has two blue resolutions and no deeper order, so its escape root solves $2\varepsilon_3 = \mu$, i.e. $\varepsilon_3 = \mu/2$. Same μ as the root, but a different root because the N_j counts differ.

The 15 probabilities

Each tree is a product over its clades: a reservable observed clade that keeps an observed split contributes $(1 - \mu)$ CCP; a clade that escapes contributes $\varepsilon(C)^{\# \text{novel}}$ for its blue region; cherries contribute 1.

	Newick	Category	Symbolic	Value
T_1	$((A, B), C), D)$	observed	$(1 - \mu)^2 c_{\text{obs}}$	0.39484375
T_2	$((D, C), B), A)$	observed	$(1 - \mu)^2 c_{\text{obs}}$	0.39484375
T_3	$((A, B), (C, D))$	α -expansion	$(1 - \mu) c_{\text{bal}}$	0.11875000
<i>One blue node: a 3-taxon clade over an observed cherry ($\# \text{novel} = 1$, root escapes)</i>				
4	$((A, B), D), C)$	blue $\{A, B, D\}$	$\varepsilon_{\text{root}}$	0.02336257
5	$((C, D), A), B)$	blue $\{A, C, D\}$	$\varepsilon_{\text{root}}$	0.02336257
<i>One blue node: a novel cherry under an observed triple (root stays observed, 3-clade escapes)</i>				
6	$((A, C), B), D)$	blue $\{A, C\}$	$(1 - \mu) c_{\text{obs}} \frac{\mu}{2}$	0.01039063
7	$((B, C), A), D)$	blue $\{B, C\}$	$(1 - \mu) c_{\text{obs}} \frac{\mu}{2}$	0.01039063
8	$((B, C), D), A)$	blue $\{B, C\}$	$(1 - \mu) c_{\text{obs}} \frac{\mu}{2}$	0.01039063
9	$((B, D), C), A)$	blue $\{B, D\}$	$(1 - \mu) c_{\text{obs}} \frac{\mu}{2}$	0.01039063
<i>Two blue nodes: fully novel ($\# \text{novel} = 2$, root escapes)</i>				
10	$((A, D), B), C)$	blue $\{A, D\}, \{A, B, D\}$	$\varepsilon_{\text{root}}^2$	0.00054581
11	$((B, D), A), C)$	blue $\{B, D\}, \{A, B, D\}$	$\varepsilon_{\text{root}}^2$	0.00054581
12	$((A, C), D), B)$	blue $\{A, C\}, \{A, C, D\}$	$\varepsilon_{\text{root}}^2$	0.00054581
13	$((A, D), C), B)$	blue $\{A, D\}, \{A, C, D\}$	$\varepsilon_{\text{root}}^2$	0.00054581
14	$((A, C), (B, D))$	blue $\{A, C\}, \{B, D\}$	$\varepsilon_{\text{root}}^2$	0.00054581
15	$((A, D), (B, C))$	blue $\{A, D\}, \{B, C\}$	$\varepsilon_{\text{root}}^2$	0.00054581
sum			1	1.00000000

Values shown at the fitted $\alpha = 0.40$, $\mu = 0.05$, for which $c_{\text{obs}} = \frac{7}{16} = 0.4375$, $c_{\text{bal}} = \frac{1}{8} = 0.125$, and $\varepsilon_{\text{root}} = \frac{\sqrt{1.3}-1}{6} = 0.0233625708 \dots$

Why it sums to exactly 1

Group the fifteen terms by the three building blocks:

$$\begin{aligned}
 \sum_{\text{all } 15} &= \underbrace{2(1 - \mu)^2 c_{\text{obs}}}_{T_1, T_2} + \underbrace{(1 - \mu) c_{\text{bal}}}_{T_3} + \underbrace{4(1 - \mu) c_{\text{obs}} \frac{\mu}{2}}_{\text{trees 6-9}} + \underbrace{2\varepsilon_{\text{root}} + 6\varepsilon_{\text{root}}^2}_{\text{trees 4, 5, 10-15}} \\
 &= (1 - \mu) c_{\text{obs}} [2(1 - \mu) + 2\mu] + (1 - \mu) c_{\text{bal}} + (2\varepsilon_{\text{root}} + 6\varepsilon_{\text{root}}^2) \\
 &= (1 - \mu)(2c_{\text{obs}} + c_{\text{bal}}) + \mu = (1 - \mu) \cdot 1 + \mu = 1.
 \end{aligned}$$

The escaped mass $2\varepsilon_{\text{root}} + 6\varepsilon_{\text{root}}^2$ is, by definition of $\varepsilon_{\text{root}}$, exactly the reserve μ ; and the backbone CCPs satisfy $2c_{\text{obs}} + c_{\text{bal}} = 1$. So the distribution is normalised for any $\alpha \in (0, \infty)$ and $\mu \in (0, 1)$, with no

leftover reserve. (This holds under `TailMode.NONE`. The optimiser's default `TailMode.BOUND` subtracts a strict upper bound on the deeper-than- k reserve tail, which is empty here, so it sub-normalises to 0.99975; on a finite enumerated space `NONE` is exact.)